Question paper (build 07.05.20.19)

INSTRUCTIONS

Please read the instructions carefully before doing the questions.

- You can use materials in your computer, notebook and text book.
- You are **NOT allowed** to use any device to share data with others.

Beside the above conditions, students must follow the following requirements:

1. **The work must complete by using Visual Studio 2022++**
2. **The Framework must be .NET 8.0**
3. **THIS PART IS VERY IMPORTANT, PLEASE READ IT CAREFULLY AND FOLLOW THE INSTRUCTIONS.**
    - You are given a database script (.sql file) in Zip file. Execute the script before doing questions.
    - **You must use the solution in received Given Materials folder.**
    - **You are not allowed to add any libraries via NuGet Package Manager.**
    - **Just one of above requirements is violated, your work will be considered as invalid.**

On completion, submit the **whole solution folder.**

Before submitting, you can delete the folder [bin] in each project to reduce the size of the solution, to fit the requirements of PEA_Client.

Zoom - + 105% Close

Search 36% 4:12 PM 7/13/2026

Question paper (build 07.05.20.19)

Question 1. (4 points)

You are asked to write a console application that includes public **Student** and **Course** classes.

The **Student** class contains:

- Two **public properties** (not public fields): StudentID (int) and StudentName (string).
- Necessary constructors (The code on line 13, 16, 17, 25 of function Main below must work).
- And other methods if you need.

The **Course** class contains:

- Two **public properties** (not public fields): **CourseID**(int), **CourseTitle**(string), the **private field** (not public) has type **Dictionary<Student, double>**.

  The Dictionary<Student, double> cover the list of students participating in the course and their GPA in this course.
- Necessary constructors (The code on line 14 of function Main below must work).
- Two public method:
    - void **AddStudent**(Student p, double g) to add student *[p]* with GPA as *[g]* to the course.
    - Void **RemoveStudent**(int StudentID) to remove student has id that equal [StudentID] from the course.

Zoom - + 95% Close

Search 36% 4:13 PM 7/13/2026

Question paper (build 07.05.20.19)

- The public override function **ToString**() to return string that present all information about the course: id, title, list of students and their GPA. (The code on line 18, 21, 27 of function Main below must work).
- The public event **OnNumberOfStudentChange** that is raised when the number of students of the course changes (it means when adding or removing student).

With the main function given in Figure 1 below, your code must have the same result as the Figure 2.

(You can find this code in the Given Materials.)

```
7   static void Notify(int oldNumber, int newNumber)
8   {
9       Console.WriteLine($"Number of student has changed from {oldNumber} to {newNumber}.");
10  }
    0 references
11  static void Main(string[] args)
12  {
13      Student s = new Student(1, "Trung");
14      Course c = new Course(1, "PRN211_Sum21");
15      c.AddStudent(s, 7.5);
16      c.AddStudent(new Student(2, "Hoa"), 7.8);
17      c.AddStudent(new Student(3, "Vinh"), 7.4);
18      Console.WriteLine(c);
19      c.RemoveStudent(2);
20      Console.WriteLine("\n---------After remove:");
21      Console.WriteLine(c);
22
23      Console.WriteLine("\n---------After add event handler:");
24      c.OnNumberOfStudentChange += Notify;
25      c.AddStudent(new Student(4, "Hoang Anh"), 8);
26      c.RemoveStudent(1);
27      Console.WriteLine(c);
28      Console.ReadLine();
29  }
```

Figure 1: Main function

The output should be the same as below

Zoom - + 95%

Close

Search

36% 4:13 PM 7/13/2026

Question paper (build 07.05.20.19)

```
C:\Users\lephu\source\repos\PE_PRN211_Sum21\PE_PRN211_
Course: 1 - PRN211_Sum21
Student: 1 - Trung - Mark: 7,5
Student: 2 - Hoa - Mark: 7,8
Student: 3 - Vinh - Mark: 7,4

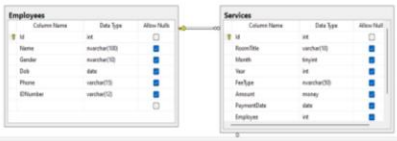

---------After remove:
Course: 1 - PRN211_Sum21
Student: 1 - Trung - Mark: 7,5
Student: 3 - Vinh - Mark: 7,4


---------After add event handler:
Number of student has changed from 2 to 3.
Number of student has changed from 3 to 2.
Course: 1 - PRN211_Sum21
Student: 4 - Hoang Anh - Mark: 8
Student: 3 - Vinh - Mark: 7,4
```

Figure 2: Output screen

Question 2. (5 points)

Use the following database diagram for doing this question.



Zoom - + 95%

Close

Search

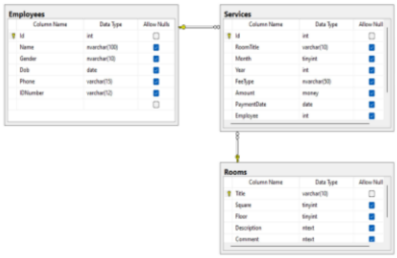
36% 4:13 PM 7/13/2026

Question paper (build 07.05.20.19)

Figure 2: Output screen

Question 2. (5 points)

Use the following database diagram for doing this question.



Note that: In the table Services, if one service has been paid it will have a date and employee to receive payment, stored in two attributes PaymentDate and Employee, otherwise these attributes are null.

In this question, you are asked to write a WPF application that allow to manage emplyee's information in database.

Note:

- 0 will be given to the work that not using database connection string in the file **appsettings.json**

Zoom

- + 95%

Close

Search

4:13 PM
7/13/2026

Question paper (build 07.05.20.19)

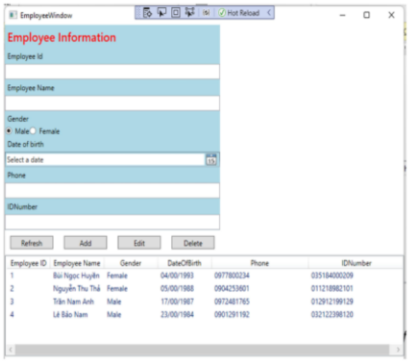


Figure 1 - Required form

- The form is designed as shown in figure above.

  The radio button [Male] is selected by default.
- List all employees that has phone number in database like the Figure 1.
- When user click on a row in the ListView, load infomation of selected Employee into panel [Employee Information] above.
- The user can click on button [Refresh] to set all data in panel [Employee Information] to default value.

Zoom

- + 95%

Close

Search

4:14 PM
7/13/2026

Question paper (build 07.05.20.19)

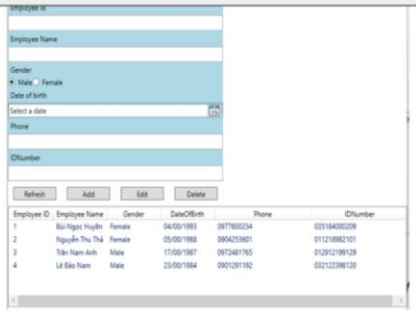


Figure 1 - Required form

- The form is designed as shown in figure above.

  The radio button [Male] is selected by default.
- List all employees that has phone number in database like the Figure 1.
- When user click on a row in the ListView, load infomation of selected Employee into panel [Employee Information] above.
- The user can click on button [Refresh] to set all data in panel [Employee Information] to default value.
- Allow the user to use panel [Employee Information] and buttons [Add], [Edit], [Delete] to create new employee and update/ delete the selected employee.

  Note that: the EmployeeID cannot be updated.

Zoom

- + 95%

Close

Search

36%

4:14 PM
7/13/2026